

SPACE SAFARI

A Journey Through Our Cosmic Neighborhood



Paul Belfrage

To my wife and children

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About This Preview

This preview contains the complete current versions of Chapter 1 — The Sun and the Edge of Home and Chapter 17 — The Stellar Life Cycle in Orion. Together they show the journey's range: from the star that shapes our immediate surroundings to a nearby region where new stars and planetary systems are forming.

About the Cover

The cover is an artistic interpretation of the Cat's Eye Nebula, also known as NGC 6543. It combines the nebula's faint outer halo with its bright, intricate centre. The inner region has been enlarged in relation to the surrounding halo for artistic effect, so the image should not be read as a scale-accurate astronomical observation.

Despite its name, a planetary nebula has nothing to do with planets. The term comes from early telescopes, in which some of these small, rounded objects looked vaguely planet-like. A planetary nebula forms late in the life of a low- or intermediate-mass star. As the star sheds its outer layers, its exposed hot core illuminates and ionizes the expanding gas, making it glow. The remaining core eventually becomes a white dwarf.

Preface

This book began as a question that stayed with me for many years: Why does the universe so often feel fragmented when we try to learn about it? There are excellent books about the solar system, others about stars, galaxies, or cosmology. But rarely does one invite the reader on a continuous journey — a guided passage outward, from the familiar to the vast, from the warmth of the Sun to the cold architecture of the cosmic web. I wanted to write the book I had once searched for: a popular science work that feels less like a catalogue of facts and more like a voyage — a kind of cosmic safari.

Space Safari is written as one continuous journey through our cosmic neighborhood, with each chapter opening onto a new but connected landscape.

I also wrote this book because I felt something was missing. Many popular science books stop at the boundary of the solar system or treat everything beyond it as an abstract blur of distant galaxies and equations. Yet the space between stars — the interstellar medium, nearby stellar groups, local bubbles and arms of the Milky Way — is rich, structured, and deeply important to understanding where we are. This book spends time there. It lingers in the regions that are often passed over, because those regions are part of our address in the universe.

My own fascination with space began early. When I was six years old, I asked my parents for a book called *Stars and Planets* by Günter D. Roth. It was far too advanced for a child who had only just learned to read properly. I could not understand most of the text, and the numbers might as well have belonged to another language. But I studied it anyway.

Years later, I still have that book. When I hold it in my hands now, I can see where it mattered most. The pages are worn, the corners softened. One spread in particular stands out — the pages showing the constellation Orion. The gutter where the two pages meet is cracked, the paper weakened by countless openings. That was the place I returned to again and again, tracing stars with my fingers, trying to imagine depth, distance, and scale long before I had the words for them. This book is written in the spirit of that cracked spine.

It does not assume that curiosity must wait for complete understanding. Instead, it invites the reader to look first, wonder first, and then slowly build understanding. Concepts are introduced where they belong in the journey, not all at once. The goal is not to overwhelm, but to orient — to give the reader a sense of where they are at every stage of the voyage.

The universe is vast, but it is not random. It has structure, history, and continuity. The atoms in our bodies were forged in stars. The planets beneath our feet are shaped by the same forces that sculpt distant worlds. Even the most remote galaxies are connected to us by light that has traveled for billions of years to reach our eyes. When we explore space, we are not escaping Earth — we are learning how deeply we belong to the cosmos.

Preface

Space Safari is written for readers who are curious but not necessarily specialized; for those who want accuracy without abstraction, depth without detachment. It is meant to be read slowly, chapter by chapter, as a guided walk through the universe — pausing often, looking around, and occasionally looking back toward home.

If this book succeeds, it will not merely inform. It will leave the reader with a sense of orientation — a feeling of knowing where we are, how we arrived here, and how far the journey continues beyond the horizon of the familiar.

The universe is not something we observe from the outside. We are inside it. This book is an invitation to explore that fact.

Paul Belfrage

July 2026

Introduction

Every journey begins with orientation. Before we move, we need to know where we are.

We live on a small rocky planet orbiting an ordinary star. That star is one among hundreds of billions in a spiral galaxy, which itself drifts through a universe structured on scales almost too large to imagine. This is our address in the cosmos — not a poetic idea, but a physical reality defined by distance, motion, gravity, and time.

Space Safari is a guided journey through that reality.

The universe is often presented to us all at once: planets beside galaxies, black holes beside atoms, distances collapsing into a single overwhelming image. But the universe is not experienced all at once. It is layered. It is hierarchical. Meaning emerges not from scale alone, but from relationships — how objects are connected, how they influence one another, and how structure builds from the small to the immense.

This book moves outward step by step, beginning at home.

We start with the Sun, because everything in our immediate cosmic environment depends on it. From there, we travel outward through the Moon and the inner planets, across the debris and boundaries of the solar system, and into the space between stars. We then widen our view to nearby stellar neighborhoods, the structure of the Milky Way, and eventually to the vast patterns that define the universe on its largest scales. Each stage of the journey builds on the one before it, not by increasing complexity for its own sake, but by expanding context.

The phrase cosmic neighborhood is used deliberately throughout this book. It does not refer to the entire universe, nor does it imply closeness in a human sense. Instead, it describes the regions of space that are physically, historically, and gravitationally connected to us. These are the structures that shaped the matter we are made of, the forces that govern our motion through space, and the environments that define where we can exist.

You will not need to solve equations, and no prior knowledge of physics or astronomy is required. Concepts are introduced where they become relevant, and technical language is used only when it clarifies rather than obscures. The aim is not to compress the universe into a list of facts, but to unfold it gradually, allowing scale and structure to become clear.

As the journey progresses, one idea returns again and again: balance. Planets survive or fail based on delicate interactions between heat, atmosphere, gravity, and time. Stars shape their surroundings through light and motion. Galaxies grow not in isolation, but through interaction and structure. Even the largest patterns in the universe emerge from simple rules applied over immense spans of time.

Introduction

Understanding space, then, is not about memorizing distances or names. It is about learning how systems behave — and how small changes can lead to vastly different outcomes.

Throughout the book, you will encounter familiar objects seen in unfamiliar ways, and distant structures described not as abstractions, but as parts of a continuous environment. You are encouraged to pause, to imagine, and occasionally to look back toward Earth — not as a point of reference to escape from, but as the place that makes all perspective possible. This is not a book about leaving Earth behind. It is a book about understanding where Earth belongs.

By the time you reach the outermost chapters, the goal is not that you remember every detail, but that you carry with you a clearer sense of orientation — a feeling for scale, connection, and continuity. The universe will still be vast. It will still be strange. But it will feel less distant, and more structured, than before.

In parallel with this book, there is a companion website: spacesafari-thebook.com. The site extends the journey beyond the printed page, offering curated links, visual material, and optional deep dives for readers who want to explore further. It is not required reading, and it does not repeat the content of the book. Instead, it provides additional context, updated resources, and pointers to observations, missions, and discoveries connected to each stage of the journey. The book stands on its own, but the website is there for those who wish to linger a little longer at each stop along the way.

The journey begins close to home.

We start with the star that makes all of this possible.

Chapter 1 — The Sun and the Edge of Home

Our Star

Our journey begins with the object that governs the solar system. The Sun is a sphere of hot plasma about 1.4 million kilometers wide and lies, on average, about 150 million kilometers from Earth. That distance is one astronomical unit, or AU; light crosses it in about eight minutes. The Sun's gravity holds the planets, dwarf planets, asteroids, and comets in orbit. Its radiation supplies most of the energy that drives Earth's climate and biosphere (NASA Science 2025).

Deep in the core, fusion combines hydrogen nuclei into helium. About 600 million tonnes of hydrogen take part each second. The resulting helium has slightly less mass than the original particles; roughly four million tonnes of mass are converted into energy, mostly emerging eventually as radiation and neutrinos. The numbers are huge, but the Sun is larger still: this steady loss of mass is tiny compared with the Sun's total mass.

Energy does not travel straight from the core to the surface. In the dense interior, photons are repeatedly absorbed and re-emitted. Models estimate that energy takes on the order of 170,000 years to pass through the radiative zone, although an individual photon does not survive that journey unchanged. Once light leaves the visible surface, it reaches Earth in about eight minutes (NASA Science 2025).

Inside the Sun

The Sun is not solid. Its matter is mainly plasma: gas in which many atoms have separated into charged nuclei and electrons. Gravity pulls this material inward while pressure produced by the hot interior pushes outward. Their balance keeps the Sun stable over long periods.

Source: NASA (public domain), via Wikimedia Commons

The core reaches about 15 million kelvins. Its temperature and pressure allow the proton-proton chain to operate. In this chain of reactions, hydrogen nuclei are gradually fused into helium. The process releases energy because the final helium nucleus has slightly less mass than the particles that went into it. The missing mass becomes energy, while neutrinos escape almost directly from the core and help scientists test solar models (Vinyoles et al. 2017).

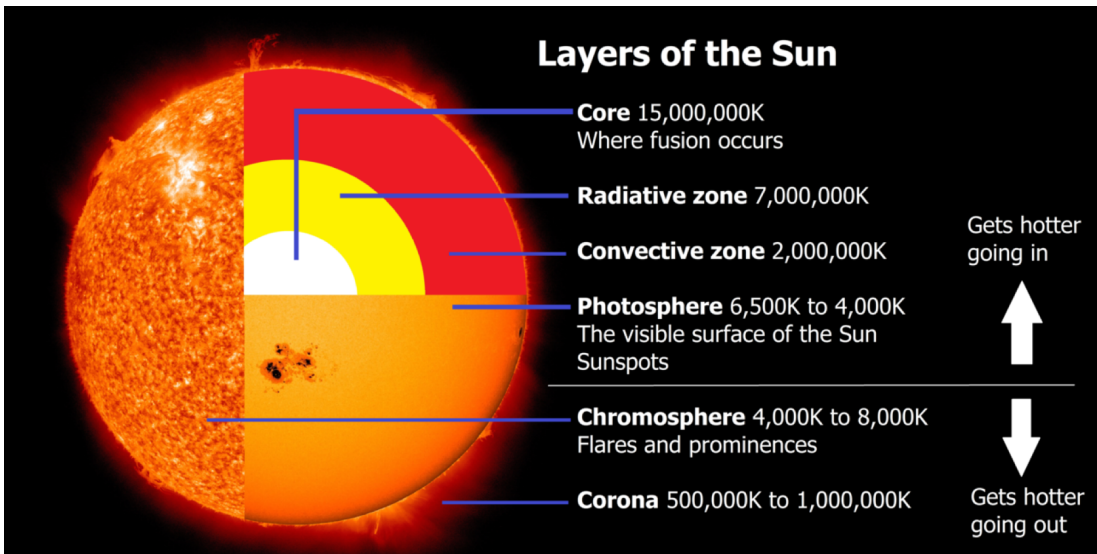


Figure 1: Layers of the Sun.

The radiative zone carries energy outward mainly through radiation. A photon does not fly freely across this region like a beam through empty space. The gas is so dense that photons are absorbed, re-emitted, and scattered again and again by charged particles. Each step may be tiny and in a random direction, so the energy slowly wanders outward instead of taking a straight path.

The convection zone begins where rising and sinking plasma becomes the more efficient way to transport energy. This motion produces the granular pattern visible at the surface.

The photosphere is the layer from which most visible sunlight escapes. It is roughly 400 kilometers thick and has an effective temperature near 5,800 kelvins. Sunspots appear darker because they are cooler than their surroundings, though they remain extremely hot.

The chromosphere and corona form parts of the solar atmosphere. Temperature describes the average motion energy of particles: hotter particles move faster. By that measure, the corona can reach one to several million kelvins, far hotter than the photosphere below it. But the corona is also extremely thin, so it contains little material compared with the dense visible surface. A spacecraft or particle in the corona would meet fast-moving particles, but not many of them. Magnetic fields, waves, turbulence, and many small releases of magnetic energy are all thought to contribute to this heating. Scientists understand important parts of the process, but no single explanation yet accounts for every observation (Klimchuk 2015).

Reading Sunlight

Sunlight appears white because it contains a broad range of visible wavelengths. A prism separates those wavelengths into colors. Shorter-wavelength photons carry more energy than longer-wavelength photons, while the overall distribution of sunlight provides information about the photosphere's temperature.

Dark absorption lines cross the solar spectrum. They form when atoms and ions absorb particular wavelengths. Each chemical element produces a characteristic pattern, allowing astronomers to identify hydrogen, helium, sodium, calcium, iron, and many others. Modern studies combine these lines with models of the solar atmosphere to estimate composition; those estimates continue to be refined (Asplund et al. 2021).

Spectra reveal more than composition. Small shifts and changes in spectral lines help researchers measure motion, temperature, density, and magnetic fields. Because the Sun can be observed in far greater detail than other stars, it provides a laboratory for testing stellar physics.

Magnetism and Space Weather

The Sun's magnetic field is generated by moving, electrically charged plasma inside the star. The Sun also rotates at different rates: faster near the equator than near the poles. Together, internal flows and differential rotation help maintain a changing magnetic field through a process called the solar dynamo.

Sunspots, flares, and coronal mass ejections are connected to this magnetic activity. A solar flare releases electromagnetic radiation. A coronal mass ejection sends a large cloud of magnetized plasma into space. The two phenomena can occur together, but they are not the same event.

If a strong eruption is directed toward Earth, it can disturb the magnetosphere and upper atmosphere. Possible effects include brighter auroras, degraded radio communication, changes to satellite operations, and electrical currents in power networks. The consequences depend on the eruption and the orientation of its magnetic field.

Sunspot numbers usually rise and fall over about eleven years. The timing and strength differ from one cycle to the next, and the Sun's global magnetic polarity reverses near solar maximum. The full magnetic pattern therefore takes roughly twenty-two years to return to its previous orientation (Hathaway 2015).

Source: NASA Goddard Space Flight Center (public domain), via Wikimedia Commons

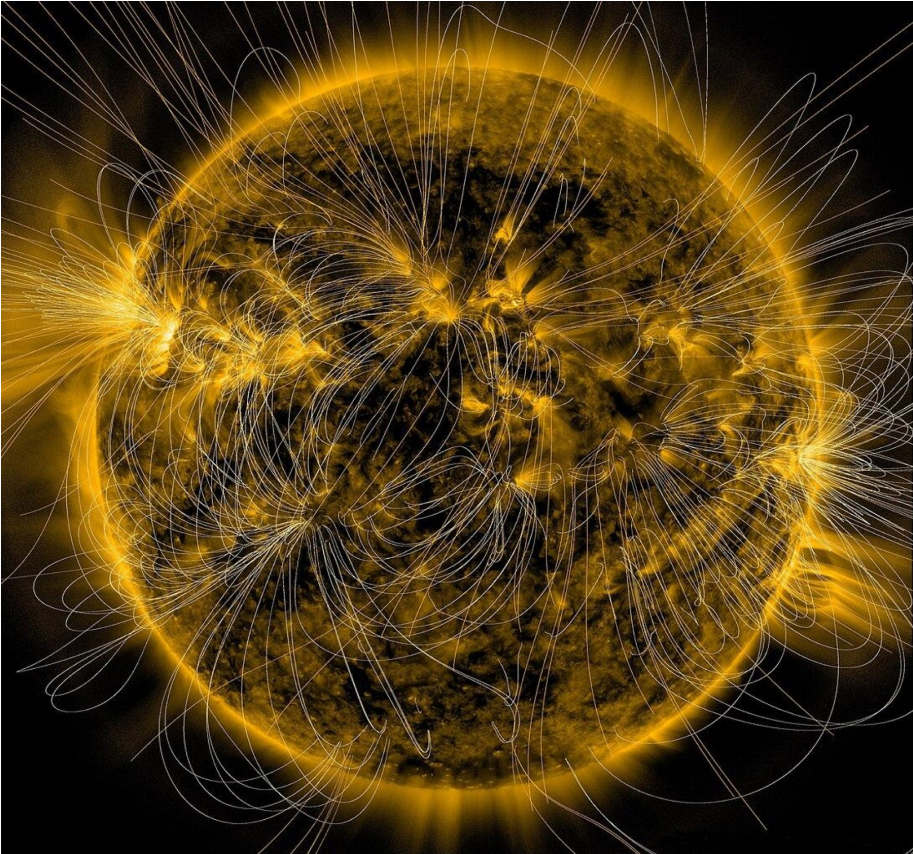


Figure 2: Solar activity observed by NASA's Solar Dynamics Observatory.

Is the Sun an Ordinary Star?

The Sun is classified as a G2 V star. “G2” describes its spectrum and surface temperature, while “V” identifies it as a main-sequence star that produces energy by fusing hydrogen in its core.

Calling it an “ordinary” or “average” star can be misleading. The Sun is neither a rare giant nor an exceptionally massive star, but most stars in the Milky Way are smaller and cooler red dwarfs. The Sun is therefore more massive and luminous than the majority of stars, even though its behavior is well within the normal main-sequence range.

Its properties have supported long-term habitability on Earth, but that does not mean only Sun-like stars can host life. Red dwarfs can produce frequent flares, and planets in their close habitable zones may become tidally locked. Yet an atmosphere or ocean could move heat from the day side to the night side, so tidal locking does not automatically create one scorched hemisphere and one frozen hemisphere. The habitability of such worlds remains an open research question (Shields et al. 2016).

Massive blue stars have much shorter lives, leaving less time for slow biological evolution of the kind that occurred on Earth. We do not know, however, the minimum time required for life to begin elsewhere. The responsible conclusion is that the Sun has proved suitable for Earth, not that it is the only kind of star that could support life.

How We Observe the Sun

Telescopic observations of sunspots in the early seventeenth century showed that the Sun changes. Spectroscopy later revealed its chemical fingerprints, and twentieth-century nuclear physics explained its main energy source.

Modern missions observe different parts of the Sun and its environment. The Solar and Heliospheric Observatory (SOHO) has monitored the Sun since 1995. NASA’s Parker Solar Probe passed only about 6.1 million kilometers above the visible surface on 24 December 2024, the closest approach by any spacecraft (NASA Science 2024). ESA’s Solar Orbiter used its inclined orbit to obtain the first clear high-angle images of the Sun’s south pole in March 2025 (European Space Agency 2025). Together with observatories near Earth, these missions trace magnetic fields, the solar wind, energetic particles, and changes across the solar cycle.

The Sun’s Past and Future

The Sun formed about 4.6 billion years ago when part of a cloud of gas and dust collapsed. Most material collected in the center; the surrounding disk supplied the matter from which planets and smaller bodies developed.

The Sun is a little less than halfway through its main-sequence lifetime. It has not remained perfectly unchanged: stellar models indicate that its luminosity has gradually increased since its youth. It should continue fusing hydrogen in its core for roughly another five billion years.

After core hydrogen is depleted, the Sun will pass through several stages and eventually expand into a red giant. Mercury and Venus are expected to be engulfed. Earth's fate depends on competing effects, including the Sun's loss of mass and tidal interaction with its extended atmosphere; some models predict that Earth will also be engulfed (Schröder and Smith 2008). Long before then, increasing solar luminosity will make Earth's present surface conditions impossible.

The Sun will later lose much of its outer material. That gas may become an ionized shell resembling a faint planetary nebula, although the exact brightness and structure are model-dependent. The name is misleading: a planetary nebula is not connected with planets. Early astronomers chose the term because some of these small, rounded objects looked vaguely like planets through their telescopes.

A planetary nebula forms when a low- or intermediate-mass star sheds its outer layers near the end of its life. The exposed hot core sends ultraviolet radiation into the expanding gas, causing it to glow. The gas can be shaped into shells, rings, knots, and jets by stellar winds, magnetic fields, and possibly a companion star. The remaining core becomes a white dwarf and then cools extremely slowly over times longer than the present age of the universe.

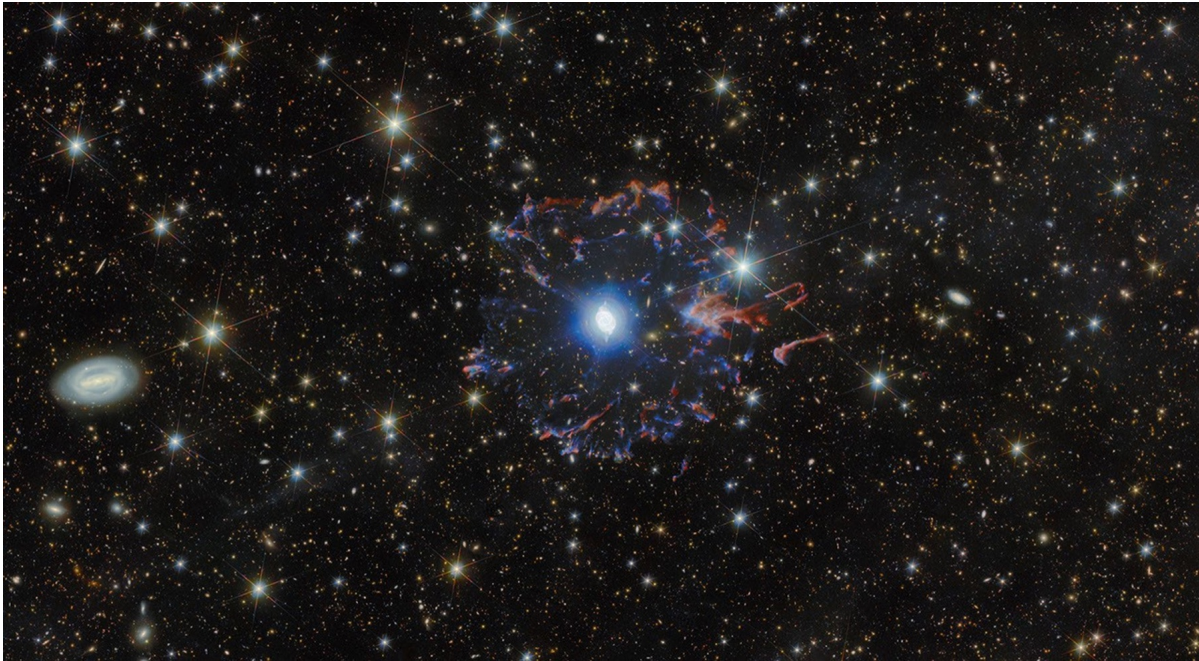


Figure 3: The Cat's Eye Nebula, NGC 6543, seen by ESA's Euclid telescope with high-resolution detail in the centre supplied by Hubble. The wide view reveals the faint outer halo of gas expelled before the brighter central nebula formed.

Source: ESA/Hubble & NASA, ESA Euclid/Euclid Consortium/NASA/Q1-2025, J.-C. Cuillandre & E. Bertin (CEA Paris-Saclay), and Z. Tsvetanov. CC BY-SA 3.0 IGO, via NASA Science.

The Sun and Earth

Earth's daily cycle comes from its rotation, not from the Sun changing between day and night. The seasons arise mainly from Earth's axial tilt, which changes the angle and duration of sunlight during the year. Earth is actually closest to the Sun during Northern Hemisphere winter, showing why distance is not the main cause of the seasons.

Solar energy drives photosynthesis, weather, and much of the circulation of the atmosphere and oceans. Most food chains ultimately depend on sunlight captured by plants, algae, and cyanobacteria. Fossil fuels store chemical energy originally captured by ancient organisms.

The Sun also changes over shorter and longer periods. Its eleven-year activity cycle affects the upper atmosphere and near-Earth space, while its luminosity evolves gradually over billions of years. Earth's climate is also shaped by the atmosphere, oceans, ice, land, orbital variations, geological processes, and human activity. The Sun is the main external energy source, but it is not the only factor controlling climate.

Summary

The Sun is a G2 V main-sequence star whose gravity and radiation dominate the solar system. Fusion in its core converts hydrogen into helium, while energy moves outward through radiative transfer and convection before escaping from the photosphere.

Moving plasma generates a changing magnetic field. Sunspots, flares, coronal mass ejections, and the solar wind create space weather that can affect Earth and technology. Missions including SOHO, Parker Solar Probe, and Solar Orbiter observe these processes from different viewpoints.

The Sun is more massive and luminous than most stars, which are red dwarfs, but it is not exceptional among main-sequence stars. It formed about 4.6 billion years ago and should remain on the main sequence for about another five billion years. Its later red-giant and white-dwarf stages are strongly supported by stellar evolution, while some details — including Earth's exact fate and the appearance of the final nebula — remain model-dependent.

Chapter 17 — The Stellar Life Cycle in Orion

Orion as a Nearby Laboratory

Orion is more than a familiar winter constellation. It contains one of the nearest large regions where astronomers can study massive stars, young stars, molecular clouds, protoplanetary disks, and stellar feedback in the same part of the sky.

At the center of this story is the Orion Molecular Cloud Complex, a large collection of gas and dust with substantial depth along our line of sight. Orion A, the part containing the Orion Nebula, is roughly 1,300 light-years away, while other clouds and young stellar groups lie farther away. The complex includes Orion A, Orion B, the Orion Nebula, the Horsehead Nebula, the Flame Nebula, and many smaller structures (Bally 2008).

Inside these clouds, gravity gathers gas into dense cores. Some cores collapse into protostars. Around many young stars, disks of gas and dust remain. Those disks are important because they can become the raw material for planets.

Source: NASA / ESA / M. Robberto / Hubble Space Telescope Orion Treasury Project Team (public domain), via Wikimedia Commons.

The Orion Nebula and the Trapezium

The Orion Nebula, also called Messier 42, is the brightest part of the Orion star-forming region. It is lit mainly by a small group of young massive stars called the Trapezium. Their ultraviolet radiation ionizes nearby hydrogen gas, making the nebula glow.

Hubble observations have revealed many protoplanetary disks in the Orion Nebula, often called proplyds (Ricci et al. 2008). Some are being eroded by radiation from nearby massive stars. Others are partly shielded inside dusty gas. JWST spectra have also confirmed externally irradiated disks around brown dwarfs in the Orion Nebula Cluster (Luhman et al. 2024). Orion therefore shows that planet-forming disks can exist in crowded, high-radiation environments, but also that such environments can be harsh.

Source: NASA/ESA, J. Bally, H. Throop, and C. R. O'Dell, Hubble Space Telescope (CC BY 4.0), via Wikimedia Commons.

Source: NASA, C. R. O'Dell and S. K. Wong (Rice University), Hubble's Visible and Infrared Views of the Trapezium, public domain.



Figure 1: The Orion Nebula (M42), a nearby star-forming region rich in gas and dust.

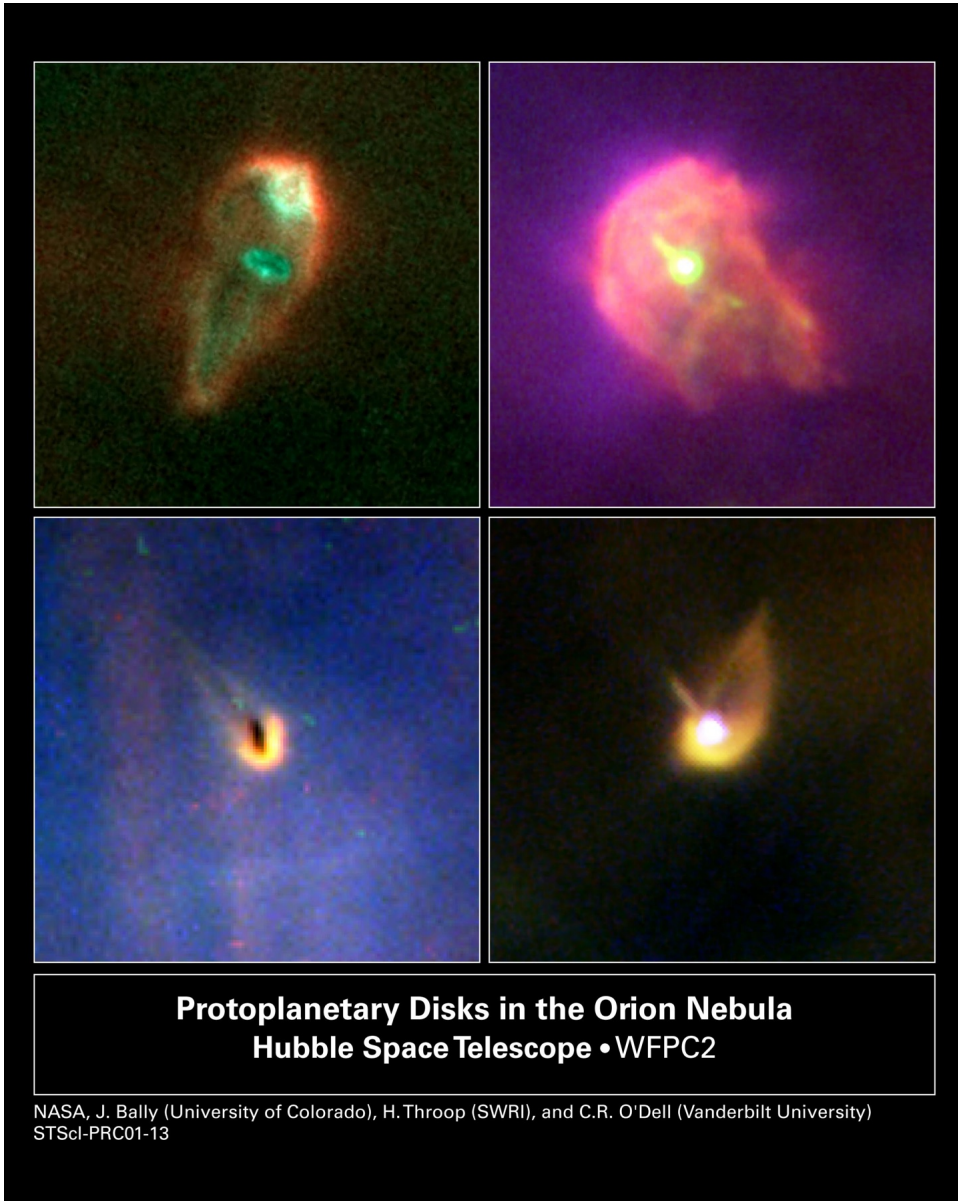


Figure 2: Four proplyds in the Orion Nebula, observed by Hubble. Each bright crescent marks gas heated by ultraviolet light from nearby massive stars; the dark or bright central region is the protoplanetary disk and its young star. Material driven from the disk forms a tail pointing away from the source of the radiation.

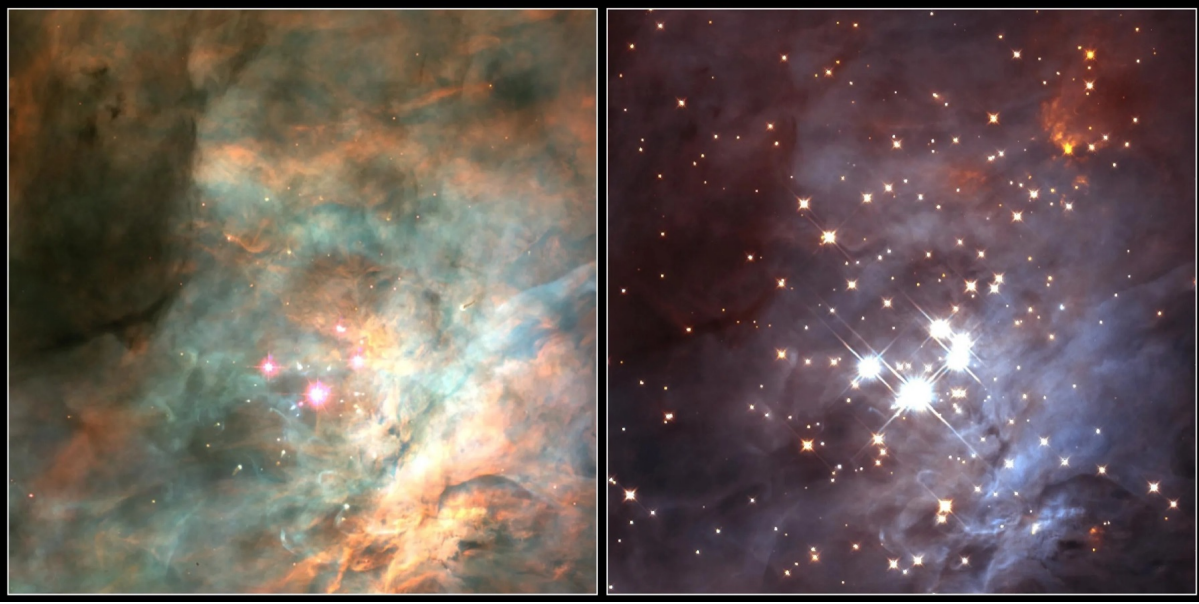


Figure 3: Trapezium in optical (left) and infrared light (right) from Hubble.

The Trapezium stars also shape the cloud around them. Their winds and radiation carve cavities and push gas outward. In some places this feedback may compress gas; in others it disperses it. Feedback is therefore not simply creative or destructive. It depends on location, density, and timing.

Massive Stars in Orion

The three stars of Orion's Belt—Alnitak, Alnilam, and Mintaka—are hot, massive stars. They are much brighter than the Sun and live much shorter lives. Massive stars burn through their nuclear fuel quickly because their cores are under extreme pressure and temperature.

These stars affect their surroundings while they are still alive. Their ultraviolet radiation ionizes gas. Their winds push against nearby clouds. Their eventual deaths can produce supernovae that return heavy elements to space and drive shock waves through the interstellar medium.

This is why massive stars matter far beyond their own lifetimes. They help regulate star formation and enrich the galaxy with elements made inside stars and during stellar explosions. But they are not gentle architects following a fixed plan. Their effects can trigger star formation in one region while shutting it down in another.

Betelgeuse and Rigel

Betelgeuse and Rigel show two very different faces of massive-star evolution.

Betelgeuse is a red supergiant. It has expanded enormously and is losing material from its outer layers. It will eventually end its life as a core-collapse supernova, but the timing is uncertain on human timescales. Its famous dimming in 2019–2020 was not a sign of an imminent explosion. It followed an unusual surface mass ejection; as the ejected gas cooled, dust formed and obscured part of the star (Dupree and Montargès 2025).

Rigel is a blue supergiant: hotter and bluer than Betelgeuse, and also a massive star with a short lifetime compared with the Sun. It too is expected to end in a dramatic late stage, but massive-star evolution depends on mass, rotation, mass loss, magnetic fields, and whether the star has companions. A simple one-line future is rarely enough.

Betelgeuse and Rigel are massive stars in different visible stages of evolution. Their contrast shows how quickly massive stars can change by astronomical standards.

Supernova Feedback Around Orion

Orion is also surrounded by large gas structures shaped by earlier generations of massive stars. Barnard’s Loop is a huge arc of ionized gas across Orion. The larger Orion–Eridanus Superbubble is a complex region shaped by winds, radiation, and likely multiple supernovae from massive stars associated with Orion (Ochsendorf et al. 2015).

These structures are not simple single-event shells. Modern studies describe nested shells and overlapping feedback episodes. That matters because it prevents a misleading picture: one explosion does not neatly create one ring and one new generation of stars. The actual interstellar medium is layered, moving, and repeatedly disturbed.

Feedback from massive stars can compress gas and may help trigger star formation in some locations. It can also clear gas away and halt star formation. Orion shows both effects.

What Orion Teaches Us

Orion gives astronomers a nearby view of several stages of stellar life:

- cold molecular gas,
- collapsing cores,
- protostars,
- protoplanetary disks,
- young massive stars,
- supergiants,
- and large feedback structures from earlier massive stars.

This does not mean the Sun was born in a region exactly like Orion. The Sun probably formed in a cluster or association, but the details of its birthplace are not known. Orion is useful because it shows processes that also operated in the Sun's past: cloud collapse, disk formation, stellar feedback, and chemical enrichment.

The elements in Earth and in living things were produced by several kinds of astrophysical process, including earlier generations of stars and stellar explosions. Orion lets us watch part of that recycling system nearby.

Summary

Orion is a practical laboratory for stellar evolution. In one region of the sky, we can study molecular clouds, young stars, protoplanetary disks, massive stars, supergiants, and feedback from earlier stellar generations.

The lesson is not that Orion is a perfect copy of the Sun's birthplace. It is that star formation and stellar death are connected.

Stars form from clouds, change those clouds, return material to space, and influence the next generation. Orion makes that cycle visible while also reminding us that each star-forming region has its own history.

Asplund, Martin, Anish M. Amarsi, and Nicolas Grevesse. 2021. 'The Chemical Make-up of the Sun: A 2020 Vision'. *Astronomy & Astrophysics* 653: A141. <https://doi.org/10.1051/0004-6361/202140445>.

Bally, John. 2008. 'Overview of the Orion Complex'. In *Handbook of Star Forming Regions, Volume i: The Northern Sky*, edited by Bo Reipurth. Astronomical Society of the Pacific.

Dupree, Andrea K., and Miguel Montargès. 2025. 'Betelgeuse, the Prototypical Red Supergiant'. *Annual Review of Astronomy and Astrophysics*. <https://arxiv.org/abs/2507.15966>.

European Space Agency. 2025. *Solar Orbiter's World-First Views of the Sun's South Pole*. https://www.esa.int/ESA_Multimedia/Images/2025/06/Solar_Orbiter_s_world-first_views_of_the_Sun_s_south_pole.

Hathaway, David H. 2015. 'The Solar Cycle'. *Living Reviews in Solar Physics* 12: 4. <https://doi.org/10.1007/lrsp-2015-4>.

Klimchuk, James A. 2015. 'Key Aspects of Coronal Heating'. *Philosophical Transactions of the Royal Society A* 373 (2042): 20140256. <https://doi.org/10.1098/rsta.2014.0256>.

Luhman, K. L., C. Alves de Oliveira, I. Baraffe, et al. 2024. 'JWST/NIRSpec Observations of Brown Dwarfs in the Orion Nebula Cluster'. *The Astrophysical Journal* 975 (2): 162. <https://doi.org/10.3847/1538-4357/ad7b19>.

- NASA Science. 2024. *NASA's Parker Solar Probe Makes History with Closest Pass to Sun*. <https://science.nasa.gov/science-research/heliophysics/nasas-parker-solar-probe-makes-history-with-closest-pass-to-sun/>.
- NASA Science. 2025. *Our Sun: Facts*. <https://science.nasa.gov/sun/facts/>.
- Ochsendorf, B. B., A. G. A. Brown, J. Bally, and A. G. G. M. Tielens. 2015. 'Nested Shells Reveal the Rejuvenation of the Orion-Eridanus Superbubble'. *The Astrophysical Journal* 808 (2): 111. <https://doi.org/10.1088/0004-637X/808/2/111>.
- Ricci, Luca, Massimo Robberto, and David R. Soderblom. 2008. 'The HST/ACS Atlas of Protoplanetary Disks in the Great Orion Nebula'. *The Astronomical Journal* 136 (5): 2136–51. <https://doi.org/10.1088/0004-6256/136/5/2136>.
- Schröder, Klaus-Peter, and Robert Cannon Smith. 2008. 'Distant Future of the Sun and Earth Revisited'. *Monthly Notices of the Royal Astronomical Society* 386 (1): 155–63. <https://doi.org/10.1111/j.1365-2966.2008.13022.x>.
- Shields, Aomawa L., Sarah Ballard, and John Asher Johnson. 2016. 'The Habitability of Planets Orbiting m-Dwarf Stars'. *Physics Reports* 663: 1–38. <https://doi.org/10.1016/j.physrep.2016.10.003>.
- Vinyoles, Núria, Aldo M. Serenelli, Francesco L. Villante, et al. 2017. 'A New Generation of Standard Solar Models'. *The Astrophysical Journal* 835 (2): 202. <https://doi.org/10.3847/1538-4357/835/2/202>.